

Program-10

Deploy an Application Using Kubernetes Deployment and Service
(Minikube)

Step 1: Initiate Minikube with specific CPU and memory

Start Minikube by allocating CPUs and memory.

```
minikube start --cpus=2 --memory=4096
```

Step 2: Enable Kubernetes Dashboard

Open the Kubernetes Dashboard to monitor the cluster.

```
minikube dashboard
```

Step 3: Create Deployment file

Create a file named **deployment.yaml**

```
nano deployment.yaml
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
```

```
matchLabels:
  app: nginx
template:
  metadata:
    labels:
      app: nginx
  spec:
    containers:
      - name: nginx
        image: nginx:latest
        ports:
          - containerPort: 80
```

Step 4: Create Service file

Create a file named **service.yaml**

```
nano service.yaml
```

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 5: Apply Kubernetes resources

Deploy the application and service.

```
kubectl apply -f deployment.yaml  
kubectl apply -f service.yaml
```

Step 6: Monitor pods via CLI

```
kubectl get pods
```

Step 7: Check nodes

```
kubectl get nodes
```

Step 8: Describe Pod (optional verification)

```
kubectl describe pod <pod-name>
```

Step 9: Check services

```
kubectl get svc
```

Step 10: Access application via Minikube

```
minikube service nginx-service
```

```
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ minikube dashboard
Enabling dashboard ...
  ■ Using image docker.io/kubernetes/dashboard:v2.7.0
  ■ Using image docker.io/kubernetes/metrics-scraper:v1.0.8
Some dashboard features require the metrics-server addon. To enable all features please run:

minikube addons enable metrics-server

Verifying dashboard health ...
Launching proxy ...
Verifying proxy health ...
Opening http://127.0.0.1:41879/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Gtk-Message: 09:52:13.329: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
^C
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ nano deployment.yaml
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ nano service.yaml
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/nginx-deployment configured
service/nginx-service created
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
kubia                                1/1      Running   4 (5m16s ago)    8d
nginx-deployment-6f9664446b-48fffd  1/1      Running   5 (5m26s ago)    20d
nginx-deployment-6f9664446b-nfcxw    1/1      Running   5 (5m26s ago)    20d
nodejs-pod                           1/1      Running   2 (5m16s ago)    6d18h
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ kubectl get nodes
NAME                STATUS    ROLES    AGE   VERSION
minikube            Ready     control-plane  21d   v1.34.0
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>          443/TCP           21d
kubia               NodePort    10.107.157.15 <none>          8080:32430/TCP    8d
nginx-deployment    NodePort    10.97.222.48  <none>          80:31057/TCP      20d
nginx-service       NodePort    10.103.48.1   <none>          80:32418/TCP      39s
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ minikube service nginx-service



| NAMESPACE | NAME          | TARGET PORT | URL                       |
|-----------|---------------|-------------|---------------------------|
| default   | nginx-service | 80          | http://192.168.49.2:32418 |



Opening service default/nginx-service in default browser...
1RV24MC106_Srusti_G_C@srusti-IdeaPad-Slin-3-151AH8:~/demo10$ Gtk-Message: 09:55:49.980: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
t load it.
```

